

Deep Learning project 3 plan

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1. Project objective

The objective of this project is to compare and analyze the performance of two different generative models in the image generation task. The team will experiment with various architectures to generate images of cats, investigate the influence of hyperparameters, and quantitatively and qualitatively assess the results. Furthermore, the project will explore the latent space through interpolation, address potential mode collapse issues, and observe the models' behavior when trained on a more diverse dataset containing multiple classes.

2. Dataset

The primary dataset used in this project will be the Cats Dataset from Kaggle. Since the images have varying resolutions, a preprocessing pipeline will be implemented to resize and crop all images to a fixed lower resolution (e.g., 64×64 pixels) to accommodate computational constraints. In the final stage of the project, the dataset will be expanded with images from the Dogs vs. Cats Kaggle dataset to see how the model handles a mix of two different animals.

3. Models

To provide a comprehensive comparison, two distinct generative approaches will be investigated:

- **Variational Autoencoder (VAE):** A stable baseline model that is relatively fast to train. It has a well-structured latent space, though it often produces slightly blurry images.
- **Deep Convolutional GAN (DCGAN):** A standard adversarial network that typically yields sharper images than VAEs but is prone to training instabilities and mode collapse.

4. Training setup

Due to the Kaggle GPU 30-hour weekly limit, the team plans to mitigate the high computational costs of generative models by training on scaled-down image resolutions and, if necessary, a randomly sampled subset of the dataset during preliminary hyperparameter searches. All models will be trained using fixed random seeds to ensure reproducibility.

Stage	Subject	Details
1	Baseline training	The 2 base models (VAE and DCGAN) will be trained on the preprocessed Cats dataset.
2	Hyperparameter investigation	The influence of key hyperparameters (such as learning rate, latent space dimension size, and batch size) will be evaluated for the selected models.
3	Latent space interpolation	For the selected model, two generated images will be chosen along with their latent matrices. Linear interpolation will be performed to generate 8 intermediate latent matrices, resulting in a sequence of 10 images to visualize latent space smoothness.
4	Mode collapse analysis	The models (especially the DCGAN) will be monitored for mode collapse. If the problem occurs, mitigation techniques (such as adding noise to the inputs or adjusting the generator/discriminator training ratio) will be explored.
5	Expanding the dataset	The best performing model architecture will be trained on the combined dataset of cats and dogs. The generated images will be visually checked to see if the model produces distinct cats and dogs, or rather a combination of both.

5. Experimentation outline

6. Evaluation metrics

Unlike classification tasks, evaluating generative models requires specialized metrics:

- **Quantitative metric:** Fréchet Inception Distance (FID) will be used to measure the similarity between the distribution of generated images and the real dataset. Lower FID scores will indicate better generation quality and diversity.
- **Qualitative assessment:** Visual inspection of the generated samples will be conducted to assess realism, sharpness, and the presence of visual artifacts.